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United States Patent  
Kind Code  
Date of Patent  
Inventor(s)

12415752  
B2  
September 16, 2025  
Lorenz; Klaus et al.

### Construction composition

#### Abstract

A construction composition comprises a) a cementitious binder comprising one or more calcium silicate mineral phases and one or more calcium aluminate mineral phases; b) optionally, an extraneous aluminate source; c) a sulfate source; d) an ettringite formation controller comprising (i) glyoxylic acid, a glyoxylic acid salt and/or a glyoxylic acid derivative; and (ii) at least one of a borate source and a carbonate source; and e) a polyol in an amount of 0.2 to 2.5 wt.-%, relative to the amount of cementitious binder a). The composition contains 0.05 to 0.2 mol of total available aluminate, calculated as  $\text{Al}(\text{OH})_{4.5}$ , from the calcium aluminate mineral phases plus the optional extraneous aluminate source, per 100 g of cementitious binder a); and the molar ratio of total available aluminate to sulfate is 0.4 to 2.0. The construction composition exhibits high early strength and sufficient open time.

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| <b>Appl. No.:</b>            | 18/023276                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Filed (or PCT Filed):</b> | August 25, 2021                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>PCT No.:</b>              | PCT/EP2021/073434                                                                                                                                                                                                                                                                                                                                                                                                                                         |

PCT Pub. No.: WO2022/043347

PCT Pub. Date: March 03, 2022

Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20230312413 A1   | Oct. 05, 2023    |

Foreign Application Priority Data

|    |          |               |
|----|----------|---------------|
| EP | 20192856 | Aug. 26, 2020 |
|----|----------|---------------|

Publication Classification

Int. Cl.: C04B7/02 (20060101); C04B24/02 (20060101); C04B24/04 (20060101); C04B28/04 (20060101); C04B28/06 (20060101); C04B28/14 (20060101); C04B28/18 (20060101); C04B103/40 (20060101)

U.S. Cl.:

CPC C04B7/02 (20130101); C04B24/02 (20130101); C04B24/04 (20130101); C04B28/04 (20130101); C04B28/06 (20130101); C04B28/14 (20130101); C04B28/188 (20130101); C04B2103/408 (20130101)

Field of Classification Search

CPC: C04B (7/02); C04B (24/02); C04B (24/04); C04B (28/04); C04B (28/06); C04B (28/14); C04B (28/188); C04B (2103/408); C04B (2111/00086)

References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name    | U.S. Cl. | CPC         |
|--------------|-------------|------------------|----------|-------------|
| 4430469      | 12/1983     | Burge et al.     | N/A      | N/A         |
| 6555683      | 12/2002     | Weichmann et al. | N/A      | N/A         |
| 11939273     | 12/2023     | Bandiera         | N/A      | C04B 22/143 |
| 11981606     | 12/2023     | Bandiera         | N/A      | C04B 24/06  |
| 2007/0074642 | 12/2006     | Amathieu         | N/A      | N/A         |

FOREIGN PATENT DOCUMENTS

| Patent No.    | Application Date | Country | CPC        |
|---------------|------------------|---------|------------|
| 602004011804  | 12/2008          | DE      | N/A        |
| 2012/133870   | 12/2011          | WO      | N/A        |
| 2017/212045   | 12/2016          | WO      | N/A        |
| WO-2019077050 | 12/2018          | WO      | C04B 14/26 |

## OTHER PUBLICATIONS

S.A. Abo-El-Enein, et al., "Early hydration characteristics of oil well cement pastes admixed with newly prepared organic admixture", HBRC Journal, Aug. 1, 2018, vol. 14, No. 2. cited by applicant  
Jochen Stark, et al., "Zement & Kalk: Der Baustoff Als Werkstoff" Jan. 1, 2000. cited by applicant  
Frank Etzler, "Particle size analysis: a Comparative Study of Various Methods", pp. 217-224, Jan. 1, 1995. cited by applicant  
Frank Etzler, "Particle size analysis: a Comparative Study of Various Methods II", pp. 278-282, vol. 14, Jan. 1, 1997, vol. 14. cited by applicant  
International Search Report for Application No. PCT/EP2021/073435, mailed on Aug. 25, 2021. cited by applicant  
International Written Opinion for No. PCT/EP2021/073435, mailed on Aug. 25, 2021. cited by applicant

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## Background/Summary

### CROSS REFERENCE TO RELATED APPLICATIONS

- (1) This application is a national stage application of International Application No. PCT/EP2021/073434, filed 25 Aug. 2021, which claims priority from European Patent Application No. 20192856.1, filed 26 Aug. 2020, both of which applications are incorporated herein by reference.
- (2) The present invention relates to a construction composition and a freshly mixed construction composition.
- (3) Prefabricated elements of concrete, such as modular building structures, are obtained by mixing materials including a cement, an aggregate, water, and a dispersant, molding in various moulds, and curing. Because the mould is repeatedly used many times, from the viewpoint of productivity and for enhancing the turnover rate of the mould, it is important for the concretes to exhibit high early strength.
- (4) WO 2012/133870 A1 discloses a method for producing a cured article of a hydraulic composition, comprising the steps of preparing a hydraulic composition by mixing glycerol, cement and water, the hydraulic composition comprising sulfate ions in a certain ratio; and aging and curing the hydraulic composition.
- (5) It is known that dispersants are added to aqueous slurries of hydraulic binders for improving their workability, i.e. kneadability, spreadability, sprayability, pumpability or flowability. Such admixtures are capable of preventing the formation of solid agglomerates and of dispersing the particles already present and those newly formed by hydration and in this way improving the workability. In order to convert the pulverulent binders into a freshly mixed processible form, substantially more mixing water is required than would be necessary for the subsequent hydration and hardening process. The voids formed in the concrete body by the excess of water, which subsequently evaporates, lead to poor mechanical strength and resistance. In order to reduce the excess proportion of water at a predetermined processing consistency and/or to improve the workability at a predetermined water/binder ratio, admixtures are used which are generally referred to as water-reducing agents or plasticizers.
- (6) Upon hydration of a cementitious system, ettringite is generated in a rapid reaction. Ettringite is

a calcium aluminum sulfate compound having the formula  $\text{Ca}_6\text{Al}_2(\text{SO}_4)_3 \cdot 32\text{H}_2\text{O}$  or alternatively  $3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{CaSO}_4 \cdot 32\text{H}_2\text{O}$ . This reaction is among others responsible for the setting and the development of early compressive strength of the cementitious composition. Ettringite forms as long needle-like crystals. The newly formed small needle-like ettringite crystals, however, tend to deteriorate the workability or flowability of the cementitious composition. In addition, ettringite contains 32 moles of water in its stoichiometric formula. This means that upon ettringite formation a significant amount of water is bound in the solid crystals. A further quantity of water is adsorbed at the newly developing ettringite surfaces. As a result, the flowability of the composition is reduced.

(7) WO 2019/077050 A1 discloses a set control composition for cementitious systems comprising a) an amine-glyoxylic acid condensate, and b) at least one of (i) a borate source and (ii) a carbonate source. The composition is said to suppress the formation of ettringite from the aluminate phases and to inhibit the crystallization of ettringite. In view of the retarding action of the set control composition, the dosage of dispersant(s) necessary to obtain a given flowability of the cementitious system can be reduced.

(8) However, the known ettringite retardation technologies were found to work well in some mix designs or applications and with certain cements or clinkers, while in others the effect is not as pronounced.

(9) The present invention therefore seeks to provide a construction composition exhibiting high early strength and sufficient open time. The advantageous effects of the invention should be readily achievable for a variety of cements with varying elemental compositions.

(10) The above problem is solved by a construction composition comprising a) a cementitious binder comprising one or more calcium silicate mineral phases and one or more calcium aluminate mineral phases; b) optionally, an extraneous aluminate source; c) a sulfate source; d) an ettringite formation controller comprising (i) glyoxylic acid, a glyoxylic acid salt and/or a glyoxylic acid derivative; and (ii) at least one of a borate source and a carbonate source, wherein the carbonate source is selected from inorganic carbonates having an aqueous solubility of 0.1 g.Math.L.sup.-1 or more, organic carbonates, and mixtures thereof; and e) a polyol in an amount of 0.2 to 2.5 wt.-%, relative to the amount of cementitious binder a); wherein the composition contains 0.05 to 0.2 mol of total available aluminate, calculated as  $\text{Al}(\text{OH})_4^-$ , from the calcium aluminate mineral phases plus the optional extraneous aluminate source, per 100 g of cementitious binder a); and the molar ratio of total available aluminate to sulfate is 0.4 to 2.0.

(11) The cementitious binder a) comprises one or more calcium silicate mineral phases and one or more crystalline calcium aluminate mineral phases.

(12) Conveniently, the mineralogical phases are herein indicated by their cement notation. The primary compounds are represented in the cement notation by the oxide varieties: C for  $\text{CaO}$ , M for  $\text{MgO}$ , S for  $\text{SiO}_2$ , A for  $\text{Al}_2\text{O}_3$ , \$ for  $\text{SO}_3$ , F for  $\text{Fe}_2\text{O}_3$ , and H for  $\text{H}_2\text{O}$ .

(13) In general, the calcium silicate mineral phases and calcium aluminate mineral phases constitute at least 90 wt.-% of the cementitious binder a). Further, the calcium silicate mineral phases preferably constitute at least 60 wt.-% of the cementitious binder a), more preferably at least 65 wt.-%, most preferably 65 to 75 wt.-%.

(14) Suitably, the calcium silicate mineral phases are selected from C3S (alite) and C2S (belite). The calcium silicate mineral phases provide primarily final strength properties.

(15) Generally, the amount of cementitious binder a) in the construction composition is at least 8 wt.-%, preferably at least 10 wt.-%, more preferably at least 15 wt.-%, most preferably at least 20 wt.-%, relative to the solids content of the construction composition.

(16) Suitably, the calcium aluminate mineral phases are selected from C3A, C4AF and C12A7, in particular C3A and C4AF.

(17) In an embodiment, the cementitious binder a) is Portland cement, in particular ordinary

Portland cement (OPC). The term “Portland cement” denotes any cement compound containing Portland clinker, especially CEM I within the meaning of standard EN 197-1, paragraph 5.2. A preferred cement is ordinary Portland cement (OPC) according to DIN EN 197-1. The phases constituting Portland cement mainly are alite (C3S), belite (C2S), calcium aluminate (C3A), calcium ferroaluminate (C4AF) and other minor phases. Commercially available OPC may either contain calcium sulfate (<7 wt.-%) or is essentially free of calcium sulfate (<1 wt.-%).

(18) According to the invention, the construction composition contains 0.05 to 0.2 mol of total available aluminate, calculated as  $\text{Al}(\text{OH})_{4.5}$ , from the calcium aluminate mineral phases plus the optional extraneous aluminate source, per 100 g of cementitious binder a). Preferably, the construction composition contains at least 0.065 mol, in particular at least 0.072 mol, of total available aluminate, per 100 g of cementitious binder a).

(19) It has been found that construction compositions containing at least 0.05 mol of total available aluminate per 100 g of cementitious binder a) exhibit optimum performance regarding open time before setting and early strength development. Otherwise, if the cementitious binder contains more than 0.2 mol of total available aluminate per 100 g of cementitious binder a), open time is shorter as early strength development is too fast.

(20) Commonly, approximate proportions of the main minerals in Portland cement are calculated by the Bogue formula which in turn is based on the elemental composition of the clinker determined, e.g., by means of X-ray fluorescence (XRF). Such methods provide the oxide composition of the elements. This means that the amount of Al is reported as  $\text{Al}_2\text{O}_3$ . It has been found that cements with apparently the same  $\text{Al}_2\text{O}_3$  content exhibit quite different properties regarding early strength and controllability by hydration control. Cement includes very different sources of Al of mineralogical nature and solubility. The present inventors have found that not all Al is available or accessible for the formation of ettringite. Only Al-containing mineral phases with adequate solubility in the aqueous environment of the cement paste participate in the formation of ettringite. Other Al-containing minerals such as crystalline aluminum oxides, e.g. corundum, do not generate aluminate in aqueous environments, due to their limited solubility. Consequently, elemental analysis alone cannot provide reliable values for available aluminate.

(21) Hence, the invention relies on the available aluminate, calculated as  $\text{Al}(\text{OH})_{4.5}$ . “Available aluminate” is meant to encompass mineral phases and Al-containing compounds that are capable of generating  $\text{Al}(\text{OH})_{4.5}$  in alkaline aqueous environments. Calcium aluminate phases, such as C3A ( $\text{Ca}_3\text{Al}_2\text{O}_6$ ), dissolve in an alkaline aqueous environment to yield  $\text{Al}(\text{OH})_{4.5}$  and  $\text{Ca}^{2+}$  ions. For the purpose of this invention, the concentration of mineral phases and Al-containing compounds that are capable of generating  $\text{Al}(\text{OH})_{4.5}$  is expressed as mol of  $\text{Al}(\text{OH})_{4.5}$  per 100 g of cementitious binder a).

(22) It is believed that the common calcium aluminate mineral phases—in contrast to crystalline aluminum oxides—are sources of available aluminate. Therefore, the amount of available aluminate in a given cementitious binder may be determined by methods capable of discriminating between the mineral phases constituting the cementitious binder. A useful method for this purpose is Rietveld refinement of an X-ray diffraction (XRD) powder pattern. This software technique is used to refine a variety of parameters, including lattice parameters, peak position, intensities and shape. This allows theoretical diffraction patterns to be calculated. As soon as the calculated diffraction pattern is almost identical to the data of an examined sample, precise quantitative information on the contained mineral phases can be determined.

(23) Generally, calcium aluminate mineral phases capable of generating  $\text{Al}(\text{OH})_{4.5}$  in alkaline aqueous environments are tricalcium aluminate (C3A), monocalcium aluminate (CA), mayenite ( $\text{C}_{12}\text{A}_7$ ), grossite ( $\text{CA}_2$ ), Q-phase ( $\text{C}_{20}\text{A}_{13}\text{M}_3\text{S}_3$ ) or tetracalcium aluminoferrite (C4AF). For practical purposes, if the cementitious binder a) is Portland cement, it generally suffices to assess the following mineral phases only: tricalcium aluminate (C3A), monocalcium

aluminate (CA), mayenite (C12A7) and tetracalcium aluminoferrite (C4AF), in particular tricalcium aluminate (C3A) and tetracalcium aluminoferrite (C4AF).

(24) Alternatively, the amount of available aluminate may be obtained by determining the total amount of Al from the elemental composition of the cementitious binder a), e.g., by XRF, and subtracting therefrom the amount of crystalline aluminum compounds not capable of generating available aluminate, as determined by XRD and Rietveld refinement. This method also takes into account amorphous, soluble aluminum compounds capable of generating available aluminate. Such crystalline aluminum compounds not capable of generating available aluminates include compounds of the melilite group, e.g., gehlenite (C2AS), compounds of the spinel group, e.g., spinel (MA), mullite ( $\text{Al}_{1.5}\text{O}_{1.5}\text{Si}_{1.5}\text{O}_{1.5}$ ), and corundum ( $\text{Al}_2\text{O}_3$ ).

(25) In one embodiment, the invention makes use of cementitious binders containing 0.05 to 0.2 mol of available aluminate from calcium aluminate mineral phases, as determined by, e.g., XRD analysis.

(26) Alternatively, if the cementitious binder a) intrinsically contains an insufficient concentration of available aluminate per 100 g of cementitious binder a), an extraneous aluminate source b) can be added. Hence in some embodiments, the construction composition contains an extraneous aluminate source b).

(27) The extraneous aluminate source b) provides available aluminate as defined above. Suitably, the extraneous aluminate source b) is selected from non-calciferous aluminate sources, such as aluminum(III) salts, aluminum(III) complexes, crystalline aluminum hydroxide, amorphous aluminum hydroxide; and calciferous aluminate sources such as high alumina cement, sulfoaluminate cement or synthetic calcium aluminate mineral phases.

(28) Useful aluminum(III) salts are aluminum(III) salts which readily form  $\text{Al}(\text{OH})_4^-$  in an alkaline aqueous environment. Suitable aluminum(III) salts include, but are not limited to, aluminum halides, such as aluminum(III) chloride, and their corresponding hydrates, amorphous aluminum oxides, aluminum hydroxides or mixed forms thereof, aluminum sulfates or sulfate-containing aluminum salts, such as potassium alum, and their corresponding hydrates, aluminum nitrate, aluminum nitrite and their corresponding hydrates, aluminum complexes such as aluminum triformate, aluminum triacetate, aluminum diacetate and aluminum monoacetate, aluminum containing metal organic frameworks, e.g. aluminum fumarate, e.g. Basolite™ A520, and M(II)-aluminum-oxo-hydrates, e.g. hydrogarnet. Aluminum(III) hydroxides may be crystalline or amorphous. Preferably, amorphous aluminum hydroxide is used.

(29) High aluminate cement means a cement containing a high concentration of calcium aluminate phases, e.g., at least 30 wt.-%. More precisely, said mineralogical phase of aluminate type comprises tricalcium aluminate (C3A), monocalcium aluminate (CA), mayenite (C12A7), tetracalcium aluminoferrite (C4AF), or a combination of several of these phases.

(30) Sulfoaluminate cement has a content of ye'elimite (of chemical formula  $4\text{CaO} \cdot 3\text{Al}_2\text{O}_3 \cdot \text{SO}_3$  or C4A3\$ in cement notation) of typically greater than 15 wt.-%.

(31) Suitable synthetic calcium aluminate mineral phases include amorphous mayenite (C12A7).

(32) The construction composition comprises a sulfate source c). The sulfate source is a compound capable of providing sulfate ions in an alkaline aqueous environment. Generally, the sulfate source has an aqueous solubility of at least 0.6 mmol.L<sup>-1</sup> at a temperature of 30° C. The aqueous solubility of the sulfate source is suitably determined in water with a starting pH value of 7.

(33) Specifically, the molar ratio of total available aluminate to sulfate is in the range of 0.4 to 2.0, preferably 0.57 to 0.8, in particular about 0.67. This means that the mixing ratios in the composition are adjusted so that the highest possible proportion of ettringite is formed from the available aluminate.

(34) As mentioned earlier, Portland cement in its commercially available form typically contains small amounts of a sulfate source. If the intrinsic amount of sulfate is unknown, it can be determined by methods familiar to the skilled person such as elemental analysis by XRF. As the sulfate source commonly used in the cement production, alkaline earth metal sulfates, alkali metal sulfates, or mixed forms thereof, such as gypsum, hemihydrate, anhydrite, arkanite, thenardite, syngenite, langbeinite, are typically crystalline, the amount thereof can also be determined by XRD. Both the intrinsic amount of sulfate and any added extraneous sulfate source are considered in the calculation of the molar ratio of total available aluminate to sulfate.

(35) In general, the extraneous sulfate source may be selected from calcium sulfate dihydrate, anhydrite,  $\alpha$ - and  $\beta$ -hemihydrate, i.e.  $\alpha$ -bassanite and  $\beta$ -bassanite, or mixtures thereof. Preferably the calcium sulfate source is  $\alpha$ -bassanite and/or  $\beta$ -bassanite. Other sulfate sources are alkali metal sulfates like potassium sulfate or sodium sulfate.

(36) It is envisaged that an additive can act as a source of both aluminate and sulfate, such as aluminum sulfate hexadecahydrate or aluminum sulfate octadecahydrate.

(37) Preferably, the sulfate source c) is a calcium sulfate source.

(38) According to the invention, the construction composition contains an ettringite formation controller. The ettringite formation controller comprises (i) glyoxylic acid, a glyoxylic acid salt and/or a glyoxylic acid derivative; and (ii) at least one of a borate source and a carbonate source. The carbonate source is selected from inorganic carbonates having an aqueous solubility of 0.1 g.Math.L.sup.-1 or more, organic carbonates, and mixtures thereof.

(39) It is believed that the component (i), i.e., glyoxylic acid, a glyoxylic acid salt and/or a glyoxylic acid derivative, in combination with borate ions or carbonate ions from component (ii), retard the formation of ettringite from the aluminate phases originating from the cementitious binder.

(40) Preferably, the (i) glyoxylic acid, glyoxylic acid salt and/or glyoxylic acid derivative is present in a total amount of 0.2 to 2 wt.-%, preferably 0.3 to 1 wt.-%, relative to the amount of cementitious binder a).

(41) Useful glyoxylic acid salts include alkali metal glyoxylates, e.g., sodium glyoxylate and potassium glyoxylate.

(42) Useful glyoxylic acid derivatives include glyoxylic acid polymers and glyoxylic acid adducts.

(43) In an embodiment, the glyoxylic acid polymer is an amine-glyoxylic acid condensate. The term "amine-glyoxylic acid condensate" is intended to mean a condensate of glyoxylic acid with a compound containing amino or amido groups reactive with aldehydes. Examples of compounds containing aldehyde-reactive amino or amido groups include urea, thiourea, melamine, guanidine, acetoguanamine, benzoguanamine and other acylguanamines and polyacrylamide.

(44) Preferably, the amine-glyoxylic acid condensate is a melamine-glyoxylic acid condensate, a urea-glyoxylic acid condensate, a melamine-urea-glyoxylic acid condensate and/or a polyacrylamide-glyoxylic acid condensate. Urea-glyoxylic acid condensates are particularly preferred. Useful amine-glyoxylic acid condensates and their manufacture are described in WO 2019/077050, incorporated by reference herein.

(45) The amine-glyoxylic acid condensates are obtainable by reacting glyoxylic acid with a compound containing aldehyde-reactive amino or amido groups. The glyoxylic acid can be used as an aqueous solution or as glyoxylic acid salts, preferably glyoxylic acid alkali metal salts.

Likewise, the amine compound can be used as salt, for example as guanidinium salts. In general, the amine compound and the glyoxylic acid are reacted in a molar ratio of 0.5 to 2 equivalents, preferably 1 to 1.3 equivalents, of glyoxylic acid per aldehyde-reactive amino or amido group. The reaction is carried out at a temperature of 0 to 120° C., preferably 25 to 105° C. The pH value is preferably from 0 to 8. The viscous products obtained in the reaction can be used as such, adjusted to a desired solids content by dilution or concentration or evaporated to dryness by, e.g., spray-drying, drum-drying, or flash-drying.

- (46) In general, the amine-glyoxylic acid condensates have molecular weights in the range of from 500 to 25 000 g/mol, preferably 1000 to 10 000 g/mol, particularly preferred 1000 to 5000 g/mol.
- (47) A useful glyoxylic acid adduct is a glyoxylic acid bisulfite adduct of formula
- (48)  $\text{X-Cat}^{\text{a}}$  wherein X is, independently of one another, selected from H or a cation equivalent Cat.sub.a wherein Cat is a cation not especially limited, but preferably selected from an alkali metal, alkaline earth metal, zinc, iron, ammonium, or phosphonium cation, or mixtures thereof and a is 1/n wherein n is the valence of the cation. The glyoxylic acid bisulfite adduct can be prepared as described in WO 2017/212045.
- (49) While glyoxylic acid and glyoxylic acid salts are solids, the glyoxylic acid condensates generally are liquids at ambient conditions. This can facilitate homogeneous distribution within the composition.
- (50) Component (ii) is at least one of a borate source and a carbonate source.
- (51) The borate source usually comprises a rapidly soluble, inexpensive, borate compound. Suitable borate sources include borax, boric acid, colemanite, and hexahydroborate.
- (52) If used, the borate source is preferably present in an amount of 0.3 to 1 wt.-%, preferably 0.3 to 0.5 wt.-%, relative to the amount of cementitious binder a).
- (53) The carbonate source may be an inorganic carbonate having an aqueous solubility of 0.1 g.Math.L.sup.-1 or more at 25° C. The aqueous solubility of the inorganic carbonate is suitably determined in water with a starting pH value of 7. It is understood that the pH value at the solubility limit is higher than the starting pH value.
- (54) In a preferred embodiment, the ettringite formation controller comprises (ii) a carbonate source. The presence of the carbonate source ensures that the mixing water is initially highly concentrated in carbonate ions. Carbonate ions are believed to adsorb onto mineral phase surfaces along with glyoxylic acid, glyoxylic acid salts and glyoxylic acid derivatives. The latter will also partly remain in the pore solution and initially prevent ettringite to be formed.
- (55) Preferably, the carbonate source is present in an amount of 0.3 to 1 wt.-%, preferably 0.3 to 0.5 wt.-%, relative to the amount of cementitious binder a).
- (56) The carbonate source may be an inorganic carbonate having an aqueous solubility of 0.1 g.Math.L.sup.-1 or more.
- (57) The “inorganic carbonate” is intended to mean a salt of carbonic acid, i.e., a salt which is characterized by the presence of a carbonate ion ( $\text{CO}_3^{2-}$ ) and/or hydrogen carbonate ion ( $\text{HCO}_3^-$ ).
- (58) In an embodiment, the inorganic carbonate may be suitably selected from alkali metal carbonates such as sodium carbonate, sodium bicarbonate, potassium carbonate or lithium carbonate, and alkaline earth metal carbonates satisfying the required aqueous solubility, such as magnesium carbonate. Further suitable inorganic carbonates include carbonates of nitrogenous bases such as guanidinium carbonate and ammonium carbonate. Sodium carbonate and sodium bicarbonate are especially preferred.
- (59) Alternatively, the carbonate source is selected from organic carbonates. “Organic carbonate” denotes an ester of carbonic acid. The organic carbonate is hydrolyzed in the presence of the cementitious system to release carbonate ions. In an embodiment, the organic carbonate is selected from ethylene carbonate, propylene carbonate, glycerol carbonate, dimethyl carbonate, di(hydroxyethyl)carbonate or a mixture thereof, preferably ethylene carbonate, propylene carbonate, and glycerol carbonate or a mixture thereof, and in particular ethylene carbonate and/or propylene carbonate. Mixtures of inorganic carbonates and organic carbonates can as well be used.
- (60) The weight ratio of component (i) to component (ii) is typically in the range of about 10:1 to about 1:10, preferably about 5:1 to about 1:5 or about 1:1 to about 1:4.
- (61) According to the invention, the construction composition contains a polyol in an amount of 0.2 to 2.5 wt.-%, relative to the amount of cementitious binder a).
- (62) It is believed that polyols such as glycerol chelate calcium ions of e.g. calcium sulfate or C3A.



As a result, calcium ion dissociation is accelerated. Chelation of calcium ions also stabilizes calcium in solution and accelerates the dissolution of calcium aluminate phases, thereby rendering aluminate from these calcium aluminate phases more accessible.

(63) While the amount of polyol can suitably be varied within the ranges above, it has been found that the optimum amount of polyol to be added to the inventive construction composition to some degree depends on the fineness of the cement clinker. As a general rule, the amount of polyol e) is 0.2 to 1 wt.-%, relative to the amount of cementitious binder a), if the Blaine surface area of the cementitious binder a) is 1500 to 4000 cm<sup>2</sup>/g, and the amount of polyol e) is more than 1 to 2.5 wt.-%, relative to the amount of cementitious binder a), if the Blaine surface area is more than 4000 cm<sup>2</sup>/g. However, additions such as fillers or supplemental cementitious materials can to some extent obscure the Blaine surface area of the clinker. The general rule above therefore applies primarily to cementitious binders containing essentially no additions such as fillers or supplemental cementitious materials. The Blaine surface area may be determined according to DIN EN 196-6.

(64) "Polyol" is intended to denote a compound having at least two alcoholic hydroxyl groups in its molecule, for example 3, 4, 5 or 6 alcoholic hydroxyl groups. Polyols having vicinal hydroxyl groups are preferred. Polyols having at least three hydroxyl groups bound to three carbon atoms in sequence are most preferred.

(65) The ability of the polyol to chelate calcium ions and thereby stabilize calcium in solution can be assessed by a calcium aluminate precipitation test. In an embodiment, the polyol, in a calcium aluminate precipitation test in which a test solution, obtained by supplementing 400 mL of a 1 wt.-% aqueous solution of the polyol with 20 mL of a 1 mol/L NaOH aqueous solution and 50 mL of a 25 mmol/L NaAlO<sub>2</sub> aqueous solution, is titrated with a 0.5 mol/L CaCl<sub>2</sub> aqueous solution at 20° C., inhibits precipitation of calcium aluminate up to a calcium concentration of 75 ppm, preferably 90 ppm.

(66) The test detects the precipitation of calcium aluminate by turbidity. Initially, the test solution is a clear solution. The clear test solution is titrated with a CaCl<sub>2</sub> aqueous solution at a constant dosage rate of, e.g., 2 mL/min, as described above. With ongoing addition of CaCl<sub>2</sub>, precipitation of calcium aluminate results in a change of the optical properties of the test solution by turbidity. The titration endpoint, expressed as the maximum calcium concentration (as Ca<sup>2+</sup>), before the onset of turbidity can be calculated from the elapsed time to the onset point.

(67) In a preferred embodiment, the polyol e) is selected from compounds consisting of carbon, hydrogen, and oxygen only and which do not contain a carbonyl group (C=O) in its molecule.

(68) It is understood that the term "carbonyl group" encompasses the tautomeric form of the C=O group, i.e. a pair of doubly bonded carbon atoms adjacent to a hydroxyl group (—C=C(OH)—).

(69) In an embodiment, the polyol is selected from monosaccharides, oligosaccharides, water-soluble polysaccharides, compounds of general formula (P-I) or dimers or trimers of compounds of general formula (P-I):

(70) ##STR00002## wherein X is

(71) ##STR00003## wherein R is —CH<sub>2</sub>OH, —NH<sub>2</sub>, n is an integer from 1 to 4, m is an integer from 1 to 8.

(72) In one embodiment, the polyol is selected from saccharides. Useful saccharides include monosaccharides, such as glucose and fructose; disaccharides, such as lactose and sucrose; trisaccharides, such as raffinose; and water-soluble polysaccharides, such as amylose and maltodextrins. Monosaccharides and disaccharides, in particular sucrose, are especially preferred.

(73) Compounds of formula (P-I) wherein X is (P-Ia) are generally referred to as sugar alcohols. Sugar alcohols are organic compounds, typically derived from sugars, containing one hydroxyl group (—OH) attached to each carbon atom. Useful sugar alcohols are mannitol, sorbitol, xylitol, arabitol, erythritol and glycerol. Among these, glycerol is particularly preferred. It is envisaged that carbonates of polyhydric alcohols such as glycerol carbonate can act as a polyol source.

(74) Compounds of formula (P-I) wherein X is (P-Ib) include pentaerythritol, and

tris(hydroxymethyl)aminomethane.

(75) Compounds of formula (P-I) wherein X is (P-Ic) include triethanolamine.

(76) Dimers or trimers denote compounds wherein two or three molecules of general formula (P-I) are linked via an ether bridge and which are formally derived from a condensation reaction with elimination of one or two molecules of water. Examples of dimers and trimers of compounds of formula (P-I) include dipentaerythritol and tripentaerythritol.

(77) In an embodiment, the construction composition further comprises f) at least one of a latent hydraulic binder, a pozzolanic binder and a filler material.

(78) For the purposes of the present invention, a “latent hydraulic binder” is preferably a binder in which the molar ratio  $(\text{CaO}+\text{MgO}):\text{SiO}_2$  is from 0.8 to 2.5 and particularly from 1.0 to 2.0. In general terms, the above-mentioned latent hydraulic binders can be selected from industrial and/or synthetic slag, in particular from blast furnace slag, electrothermal phosphorous slag, steel slag and mixtures thereof. The “pozzolanic binders” can generally be selected from amorphous silica, preferably precipitated silica, fumed silica and microsilica, ground glass, metakaolin, aluminosilicates, fly ash, preferably brown-coal fly ash and hard-coal fly ash, natural pozzolans such as tuff, trass and volcanic ash, calcined clays, burnt shale, rice husk ash, natural and synthetic zeolites and mixtures thereof.

(79) The slag can be either industrial slag, i.e. waste products from industrial processes, or else synthetic slag. The latter can be advantageous because industrial slag is not always available in consistent quantity and quality.

(80) Blast furnace slag (BFS) is a waste product of the glass furnace process. Other materials are granulated blast furnace slag (GBFS) and ground granulated blast furnace slag (GGBFS), which is granulated blast furnace slag that has been finely pulverized. Ground granulated blast furnace slag varies in terms of grinding fineness and grain size distribution, which depend on origin and treatment method, and grinding fineness influences reactivity here. The Blaine value is used as parameter for grinding fineness, and typically has an order of magnitude of from 200 to 1000  $\text{m}^2/\text{kg}$ , preferably from 300 to 500  $\text{m}^2/\text{kg}$ . Finer milling gives higher reactivity.

(81) For the purposes of the present invention, the expression “blast furnace slag” is however intended to comprise materials resulting from all of the levels of treatment, milling, and quality mentioned (i.e. BFS, GBFS and GGBFS). Blast furnace slag generally comprises from 30 to 45% by weight of CaO, about 4 to 17% by weight of MgO, about 30 to 45% by weight of  $\text{SiO}_2$  and about 5 to 15% by weight of  $\text{Al}_2\text{O}_3$ , typically about 40% by weight of CaO, about 10% by weight of MgO, about 35% by weight of  $\text{SiO}_2$  and about 12% by weight of  $\text{Al}_2\text{O}_3$ .

(82) Electrothermal phosphorous slag is a waste product of electrothermal phosphorous production. It is less reactive than blast furnace slag and comprises about 45 to 50% by weight of CaO, about 0.5 to 3% by weight of MgO, about 38 to 43% by weight of  $\text{SiO}_2$ , about 2 to 5% by weight of  $\text{Al}_2\text{O}_3$  and about 0.2 to 3% by weight of  $\text{Fe}_2\text{O}_3$ , and also fluoride and phosphate. Steel slag is a waste product of various steel production processes with greatly varying composition.

(83) Amorphous silica is preferably an X ray-amorphous silica, i.e. a silica for which the powder diffraction method reveals no crystallinity. The content of  $\text{SiO}_2$  in the amorphous silica of the invention is advantageously at least 80% by weight, preferably at least 90% by weight. Precipitated silica is obtained on an industrial scale by way of precipitating processes starting from water glass. Precipitated silica from some production processes is also called silica gel.

(84) Fumed silica is produced via reaction of chlorosilanes, for example silicon tetrachloride, in a hydrogen/oxygen flame. Fumed silica is an amorphous  $\text{SiO}_2$  powder of particle diameter from 5 to 50 nm with specific surface area of from 50 to 600  $\text{m}^2/\text{g}$ .

(85) Microsilica is a by-product of silicon production or ferrosilicon production, and likewise

consists mostly of amorphous SiO<sub>2</sub> powder. The particles have diameters of the order of magnitude of 0.1 μm. Specific surface area is of the order of magnitude of from 15 to 30 m<sup>2</sup>/g. -1.

(86) Metakaolin is produced when kaolin is dehydrated. Whereas at from 100 to 200° C. kaolin releases physically bound water, at from 500 to 800° C. a dehydroxylation takes place, with collapse of the lattice structure and formation of metakaolin (Al<sub>2</sub>Si<sub>2</sub>O<sub>7</sub>). Accordingly pure metakaolin comprises about 54% by weight of SiO<sub>2</sub> and about 46% by weight of Al<sub>2</sub>O<sub>3</sub>.

(87) Fly ash is produced inter alia during the combustion of coal in power stations. Class C fly ash (brown-coal fly ash) comprises according to WO 08/012438 about 10% by weight of CaO, whereas class F fly ash (hard-coal fly ash) comprises less than 8% by weight, preferably less than 4% by weight, and typically about 2% by weight of CaO.

(88) In case the construction composition contains a low amount of hydraulic binder (e.g. ≤10%), an alkaline activator can be further added to promote strength development. Alkaline activators are preferably used in the binder system, such alkaline activators are for example aqueous solutions of alkali metal fluorides, alkali metal hydroxides, alkali metal aluminates or alkali metal silicates, such as soluble waterglass, and mixtures thereof.

(89) For the purposes of the present invention, a “filler material” can be for example silica, quartz, sand, crushed marble, glass spheres, granite, basalt, limestone, sandstone, calcite, marble, serpentine, travertine, dolomite, feldspar, gneiss, alluvial sands, any other durable aggregate, and mixtures thereof. In particular, the fillers do not work as a binder.

(90) In an embodiment, the ettringite formation controller additionally comprises (iii) a polycarboxylic acid or a salt thereof whose milliequivalent number of carboxyl groups is 3.0 meq/g or higher, preferably 3.0 to 17.0 meq/g, more preferably 5.0 to 17.0 meq/g, most preferably 5.0 to 14.0 meq/g, assuming all the carboxyl groups to be in unneutralized form. By the term polycarboxylic acid, as used herein, is meant a compound or polymer having two or more carboxyl groups to the molecule.

(91) In particular, the polycarboxylic acid is selected from phosphonoalkyl carboxylic acids, amino carboxylic acids, and polymeric carboxylic acids.

(92) Suitable polycarboxylic acids are low molecular weight polycarboxylic acids (having a molecular weight of, e.g., 500 or lower), in particular aliphatic polycarboxylic acids, such as oxalic acid, malonic acid, succinic acid, glutaric acid, adipic acid, pimelic acid, fumaric acid, maleic acid, itaconic acid, citraconic acid, mesaconic acid, malic acid, tartaric acid, and citric acid.

(93) Suitable phosphonoalkylcarboxylic acids include 1-phosphonobutane-1,2,4-tricarboxylic acid, 2-phosphonobutane-1,2,4-tricarboxylic acid, 3-phosphonobutane-1,2,4-tricarboxylic acid, 4-phosphonobutane-1,2,4-tricarboxylic acid, 2,4-diphosphonobutane-1,2,4-tricarboxylic acid, 2-phosphonobutane-1,2,3,4-tetracarboxylic acid, 1-methyl-2-phosphonopentane-1,2,4-tricarboxylic acid, or 1,2-phosphonoethane-2-dicarboxylic acid.

(94) Suitable amino carboxylic acids include ethylenediamine tetra acetic acid, or nitrilotriacetic acid.

(95) Suitable polymeric carboxylic acids include homopolymers of acrylic acid, homopolymers of methacrylic acid, polymaleic acid, copolymers such as ethylene/acrylic acid copolymer and ethylene/methacrylic acid copolymer; copolymers of acrylic acid and/or methacrylic acid with sulfo or sulfonate group containing monomers. In an embodiment, the sulfo or sulfonate group containing monomers are selected from the group of vinylsulfonic acid, (meth)allylsulfonic acid, 4-vinylphenylsulfonic acid or 2-acrylamido-2-methylpropylsulfonic acid (ATBS), with ATBS being particularly preferred. It is possible that one more of the before mentioned sulfo or sulfonate group containing monomers are contained in the copolymers.

(96) In general, the molecular weight of the polymeric carboxylic acids is in the range of from 1000 to 30000 g/mol, preferably 1000 to 10 000 g/mol. The molecular weight is measured by the gel

permeation chromatography method (GPC) as indicated in detail in the experimental part.

(97) Phosphonates, which comprise two or three phosphonate groups and no carboxyl groups are preferably 1-hydroxyethylidene-1,1-diphosphonic acid (HEDP), amino-tris(methylenephosphonic acid) (ATMP) or [[(2-hydroxyethyl)imino]bis(methylene)]-bisphosphonic acid and mixtures thereof. The respective chemical formulae of the preferred di- or triphosphonates are given in the following:

(98) ##STR00004##

(99) The phosphonates are retarders for cementitious systems and have the advantage that the open time is prolonged.

(100) Preferably, the ettringite formation controller additionally comprises (iv) a  $\alpha$ -hydroxy monocarboxylic acid or a salt thereof.

(101) Preferably, the a  $\alpha$ -hydroxy monocarboxylic acid or a salt thereof (iv) is present in a total amount of 0.05 to 1 wt.-%, preferably 0.05 to 0.2 wt.-%, relative to the amount of cementitious binder a).

(102) Suitable  $\alpha$ -hydroxy monocarboxylic acids or salts thereof include glycolic acid, gluconic acid, and their salts and mixtures thereof. Sodium gluconate is particularly preferred.

(103) Although not preferred, the construction composition may comprise setting accelerators as conventionally used, e.g., in repair mortars and self-levelling underlayments, such as lithium salts, in particular lithium carbonate or lithium sulfate. It is an advantageous feature of the invention that the early strength development of the construction composition is such that lithium setting accelerators can be dispensed with. Hence, in preferred embodiments, the construction composition does not contain a lithium setting accelerator. This also serves to reduce the cost of the construction composition, as lithium setting accelerators are quite costly ingredients.

(104) Preferably, the construction composition according to the invention additionally comprises at least one dispersant for inorganic binders, especially a dispersant for cementitious mixtures like concrete or mortar.

(105) Examples of useful dispersants include comb polymers having a carbon-containing backbone to which are attached pendant cement-anchoring groups and polyether side chains, non-ionic comb polymers having a carbon-containing backbone to which are attached pendant hydrolysable groups and polyether side chains, the hydrolysable groups upon hydrolysis releasing cement-anchoring groups, colloiddally disperse preparations of polyvalent metal cations, such as  $\text{Al}^{\text{sup.3+}}$ ,  $\text{Fe}^{\text{sup.3+}}$  or  $\text{Fe}^{\text{sup.2+}}$ , and a polymeric dispersant which comprises anionic and/or anionogenic groups and polyether side chains, and the polyvalent metal cation is present in a superstoichiometric quantity, calculated as cation equivalents, based on the sum of the anionic and anionogenic groups of the polymeric dispersant, sulfonated melamine-formaldehyde condensates, lignosulfonates, sulfonated ketone-formaldehyde condensates, sulfonated naphthalene-formaldehyde condensates, phosphonate containing dispersants, phosphate containing dispersants, and mixtures thereof.

(106) Preferably, the dispersant is present in a total amount of 0.08 to 0.4 wt.-%, preferably 0.15 to 0.3 wt.-%, relative to the amount of cementitious binder a).

(107) Comb polymers having a carbon-containing backbone to which are attached pendant cement-anchoring groups and polyether side chains are particularly preferred. The cement-anchoring groups are anionic and/or anionogenic groups such as carboxylic groups, phosphonic or phosphoric acid groups or their anions. Anionogenic groups are the acid groups present in the polymeric dispersant, which can be transformed to the respective anionic group under alkaline conditions.

(108) Preferably, the structural unit comprising anionic and/or anionogenic groups is one of the general formulae (Ia), (Ib), (Ic) and/or (Id):

(109) ##STR00005## wherein  $\text{R}^{\text{sup.1}}$  is H,  $\text{C}^{\text{sub.1-C.sub.4}}$  alkyl,  $\text{CH}^{\text{sub.2COOH}}$  or  $\text{CH}^{\text{sub.2CO}}-\text{X}-\text{R}^{\text{sup.3A}}$ , preferably H or methyl; X is  $\text{NH}-(\text{C}^{\text{sub.n1H.sub.2n1}})$  or  $\text{O}-(\text{C}^{\text{sub.n1H.sub.2n1}})$  with  $n_1=1, 2, 3$  or  $4$ , or a chemical bond, the nitrogen atom or the oxygen atom being bonded to the CO group;  $\text{R}^{\text{sup.2}}$  is OM,  $\text{PO}^{\text{sub.3M.sub.2}}$ , or  $\text{O}-\text{PO}^{\text{sub.3M.sub.2}}$ ;

with the proviso that X is a chemical bond if R<sup>sup.2</sup> is OM; R<sup>sup.3A</sup> is PO.sub.3M.sub.2, or O—PO.sub.3M.sub.2;

(110) ##STR00006## wherein R<sup>sup.3</sup> is H or C.sub.1-C.sub.4 alkyl, preferably H or methyl; n is 0, 1, 2, 3 or 4; R<sup>sup.4</sup> is PO.sub.3M.sub.2, or O—PO.sub.3M.sub.2;

(111) ##STR00007## wherein R<sup>sup.5</sup> is H or C.sub.1-C.sub.4 alkyl, preferably H; Z is O or NR<sup>sup.7</sup>; R<sup>sup.7</sup> is H, (C.sub.n1H.sub.2n1)—OH, (C.sub.n1H.sub.2n1)—PO.sub.3M.sub.2, (C.sub.n1H.sub.2n1)—OPO.sub.3M.sub.2, (C.sub.6H.sub.4)—PO.sub.3M.sub.2, or (C.sub.6H.sub.4)—OPO.sub.3M.sub.2, and n1 is 1, 2, 3 or 4;

(112) ##STR00008## wherein R<sup>sup.6</sup> is H or C.sub.1-C.sub.4 alkyl, preferably H; Q is NR<sup>sup.7</sup> or O; R<sup>sup.7</sup> is H, (C.sub.n1H.sub.2n1)—OH, (C.sub.n1H.sub.2n1)—PO.sub.3M.sub.2, (C.sub.n1H.sub.2n1)—OPO.sub.3M.sub.2, (C.sub.6H.sub.4)—PO.sub.3M.sub.2, or (C.sub.6H.sub.4)—OPO.sub.3M.sub.2, n1 is 1, 2, 3 or 4; and where each M independently is H or a cation equivalent.

(113) Preferably, the structural unit comprising a polyether side chain is one of the general formulae (IIa), (IIb), (IIc) and/or (IId):

(114) ##STR00009## wherein R<sup>sup.10</sup>, R<sup>sup.11</sup> and R<sup>sup.12</sup> independently of one another are H or C.sub.1-C.sub.4 alkyl, preferably H or methyl; Z<sup>sup.2</sup> is O or S; E is C.sub.2-C.sub.6 alkylene, cyclohexylene, CH.sub.2-C.sub.6H.sub.10, 1,2-phenylene, 1,3-phenylene or 1,4-phenylene; G is O, NH or CO—NH; or E and G together are a chemical bond; A is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; n2 is 0, 1, 2, 3, 4 or 5; a is an integer from 2 to 350, preferably 10 to 150, more preferably 20 to 100; R<sup>sup.13</sup> is H, an unbranched or branched C.sub.1-C.sub.4 alkyl group, CO—NH.sub.2 or COCH.sub.3;

(115) ##STR00010## wherein R<sup>sup.16</sup>, R<sup>sup.17</sup> and R<sup>sup.18</sup> independently of one another are H or C.sub.1-C.sub.4 alkyl, preferably H; E<sup>sup.2</sup> is C.sub.2-C.sub.6 alkylene, cyclohexylene, CH.sub.2-C.sub.6H.sub.10, 1,2-phenylene, 1,3-phenylene, or 1,4-phenylene, or is a chemical bond; A is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; n2 is 0, 1, 2, 3, 4 or 5; L is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; a is an integer from 2 to 350, preferably 10 to 150, more preferably 20 to 100; d is an integer from 1 to 350, preferably 10 to 150, more preferably 20 to 100; R<sup>sup.19</sup> is H or C.sub.1-C.sub.4 alkyl; and R<sup>sup.20</sup> is H or C.sub.1-C.sub.4 alkyl;

(116) ##STR00011## wherein R<sup>sup.21</sup>, R<sup>sup.22</sup> and R<sup>sup.23</sup> independently are H or C.sub.1-C.sub.4 alkyl, preferably H; W is O, NR<sup>sup.25</sup>, or is N; V is 1 if W=O or NR<sup>sup.25</sup>, and is 2 if W=N; A is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; a is an integer from 2 to 350, preferably 10 to 150, more preferably 20 to 100; R<sup>sup.24</sup> is H or C.sub.1-C.sub.4 alkyl; R<sup>sup.25</sup> is H or C.sub.2-C.sub.4 alkyl;

(117) ##STR00012## wherein R<sup>sup.6</sup> is H or C.sub.1-C.sub.4 alkyl, preferably H; Q is NR<sup>sup.10</sup>, N or O; V is 1 if Q=O or NR<sup>sup.10</sup> and is 2 if Q=N; R<sup>sup.10</sup> is H or C.sub.1-C.sub.4 alkyl; A is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; and a is an integer from 2 to 350, preferably 10 to 150, more preferably 20 to 100; where each M independently is H or a cation equivalent.

(118) The molar ratio of structural units (I) to structural units (II) varies from 1:3 to about 10:1, preferably 1:1 to 10:1, more preferably 3:1 to 6:1. The polymeric dispersants comprising structural units (I) and (II) can be prepared by conventional methods, for example by free radical polymerization or controlled radical polymerization. The preparation of the dispersants is, for example, described in EP 0 894 811, EP 1 851 256, EP 2 463 314, and EP 0 753 488.

(119) A number of useful dispersants contain carboxyl groups, salts thereof or hydrolysable groups releasing carboxyl groups upon hydrolysis. Preferably, the milliequivalent number of carboxyl groups contained in these dispersants (or of carboxyl groups releasable upon hydrolysis of hydrolysable groups contained in the dispersant) is lower than 3.0 meq/g, assuming all the carboxyl groups to be in unneutralized form.

(120) More preferably, the dispersant is selected from the group of polycarboxylate ethers (PCEs).  
(121) In PCEs, the anionic groups are carboxylic groups and/or carboxylate groups. The PCE is preferably obtainable by radical copolymerization of a polyether macromonomer and a monomer comprising anionic and/or anionogenic groups. Preferably, at least 45 mol-%, preferably at least 80 mol-% of all structural units constituting the copolymer are structural units of the polyether macromonomer or the monomer comprising anionic and/or anionogenic groups.

(122) A further class of suitable comb polymers having a carbon-containing backbone to which are attached pendant cement-anchoring groups and polyether side chains comprise structural units (III) and (IV):

(123) ##STR00013## wherein T is phenyl, naphthyl or heteroaryl having 5 to 10 ring atoms, of which 1 or 2 atoms are heteroatoms selected from N, O and S; n<sub>3</sub> is 1 or 2; B is N, NH or O, with the proviso that n<sub>3</sub> is 2 if B is N and n<sub>3</sub> is 1 if B is NH or O; A is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; a<sub>2</sub> is an integer from 1 to 300; R<sup>26</sup> is H, C.sub.1-C.sub.10 alkyl, C.sub.5-C.sub.8 cycloalkyl, aryl, or heteroaryl having 5 to 10 ring atoms, of which 1 or 2 atoms are heteroatoms selected from N, O and S; where the structural unit (IV) is selected from the structural units (IVa) and (IVb):

(124) ##STR00014## wherein D is phenyl, naphthyl or heteroaryl having 5 to 10 ring atoms, of which 1 or 2 atoms are heteroatoms selected from N, O and S; E<sup>3</sup> is N, NH or O, with the proviso that m is 2 if E<sup>3</sup> is N and m is 1 if E<sup>3</sup> is NH or O; A is C.sub.2-C.sub.5 alkylene or CH.sub.2CH(C.sub.6H.sub.5), preferably C.sub.2-C.sub.3 alkylene; b is an integer from 0 to 300; M independently is H or a cation equivalent;

(125) ##STR00015## wherein V<sup>2</sup> is phenyl or naphthyl and is optionally substituted by 1 or two radicals selected from R<sup>8</sup>, OH, OR<sup>8</sup>, (CO)R<sup>8</sup>, COOM, COOR<sup>8</sup>, SO<sub>3</sub>R<sup>8</sup> and NO<sub>2</sub>; R<sup>7A</sup> is COOM, OCH<sub>2</sub>COOM, SO<sub>3</sub>M or OPO<sub>3</sub>M<sub>2</sub>; M is H or a cation equivalent; and R<sup>8</sup> is C.sub.1-C.sub.4 alkyl, phenyl, naphthyl, phenyl-C.sub.1-C.sub.4 alkyl or C.sub.1-C.sub.4 alkylphenyl.

(126) Polymers comprising structural units (III) and (IV) are obtainable by polycondensation of an aromatic or heteroaromatic compound having a polyoxyalkylene group attached to the aromatic or heteroaromatic core, an aromatic compound having a carboxylic, sulfonic or phosphate moiety, and an aldehyde compound such as formaldehyde.

(127) In an embodiment, the dispersant is a non-ionic comb polymer having a carbon-containing backbone to which are attached pendant hydrolysable groups and polyether side chains, the hydrolysable groups upon hydrolysis releasing cement-anchoring groups. Conveniently, the structural unit comprising a polyether side chain is one of the general formulae (IIa), (IIb), (IIc) and/or (IId) discussed above. The structural unit having pendant hydrolysable groups is preferably derived from acrylic acid ester monomers, more preferably hydroxyalkyl acrylic monoesters and/or hydroxyalkyl diesters, most preferably hydroxypropyl acrylate and/or hydroxyethyl acrylate. The ester functionality will hydrolyze to (deprotonated) acid groups upon exposure to water at preferably alkaline pH, which is provided by mixing the cementitious binder with water, and the resulting acid functional groups will then form complexes with the cement component.

(128) In one embodiment, the dispersant is selected from colloidal dispersions of polyvalent metal cations, such as Al<sup>3+</sup>, Fe<sup>3+</sup> or Fe<sup>2+</sup>, and a polymeric dispersant which comprises anionic and/or anionogenic groups and polyether side chains. The polyvalent metal cation is present in a superstoichiometric quantity, calculated as cation equivalents, based on the sum of the anionic and anionogenic groups of the polymeric dispersant. Such dispersants are described in further detail in WO 2014/013077 A1, which is incorporated by reference herein.

(129) Suitable sulfonated melamine-formaldehyde condensates are of the kind frequently used as plasticizers for hydraulic binders (also referred to as MFS resins). Sulfonated melamine-formaldehyde condensates and their preparation are described in, for example, CA 2 172 004 A1, DE 44 1 1 797 A1, U.S. Pat. Nos. 4,430,469, 6,555,683 and CH 686 186 and also in Ullmann's

Encyclopedia of Industrial Chemistry, 5th Ed., vol. A2, page 131, and Concrete Admixtures Handbook—Properties, Science and Technology, 2. Ed., pages 411, 412. Preferred sulfonated melamine-formaldehyde condensates encompass (greatly simplified and idealized) units of the formula

(130) ##STR00016##

in which  $n_4$  stands generally for 10 to 300. The molar weight is situated preferably in the range from 2500 to 80 000. Additionally, to the sulfonated melamine units it is possible for other monomers to be incorporated by condensation. Particularly suitable is urea. Moreover, further aromatic units as well may be incorporated by condensation, such as gallic acid, aminobenzenesulfonic acid, sulfanilic acid, phenolsulfonic acid, aniline, ammoniobenzoic acid, dialkoxybenzenesulfonic acid, dialkoxybenzoic acid, pyridine, pyridinemonosulfonic acid, pyridinedisulfonic acid, pyridinecarboxylic acid and pyridinedicarboxylic acid. An example of melaminesulfonate-formaldehyde condensates are the Melment® products distributed by Master Builders Solutions Deutschland GmbH.

(131) Suitable lignosulfonates are products which are obtained as by-products in the paper industry. They are described in Ullmann's Encyclopedia of Industrial Chemistry, 5th Ed., vol. A8, pages 586, 587. They include units of the highly simplified and idealizing formula

(132) ##STR00017##

(133) Lignosulfonates have molar weights of between 2000 and 100 000 g/mol. In general, they are present in the form of their sodium, calcium and/or magnesium salts. Examples of suitable lignosulfonates are the Borresperse products distributed by Borregaard LignoTech, Norway.

(134) Suitable sulfonated ketone-formaldehyde condensates are products incorporating a monoketone or diketone as ketone component, preferably acetone, butanone, pentanone, hexanone or cyclohexanone. Condensates of this kind are known and are described in WO 2009/103579, for example. Sulfonated acetone-formaldehyde condensates are preferred. They generally comprise units of the formula (according to J. Plank et al., J. Appl. Poly. Sci. 2009, 2018-2024):

(135) ##STR00018##

where  $m_2$  and  $n_5$  are generally each 10 to 250,  $M_{sup.2}$  is an alkali metal ion, such as  $Na_{sup.+}$ , and the ratio  $m_2:n_5$  is in general in the range from about 3:1 to about 1:3, more particularly about 1.2:1 to 1:1.2. Furthermore, it is also possible for other aromatic units to be incorporated by condensation, such as gallic acid, aminobenzenesulfonic acid, sulfanilic acid, phenolsulfonic acid, aniline, ammoniobenzoic acid, dialkoxybenzenesulfonic acid, dialkoxybenzoic acid, pyridine, pyridinemonosulfonic acid, pyridinedisulfonic acid, pyridinecarboxylic acid and pyridinedicarboxylic acid. Examples of suitable sulfonated acetone-formaldehyde condensates are the Melcret K1L products distributed by Master Builders Solutions Deutschland GmbH.

(136) Suitable sulfonated naphthalene-formaldehyde condensates are products obtained by sulfonation of naphthalene and subsequent polycondensation with formaldehyde. They are described in references including Concrete Admixtures Handbook—Properties, Science and Technology, 2. Ed., pages 411-413 and in Ullmann's Encyclopedia of Industrial Chemistry, 5th Ed., vol. A8, pages 587, 588. They comprise units of the formula

(137) ##STR00019##

(138) Typically, molar weights ( $M_w$ ) of between 1000 and 50 000 g/mol are obtained. Furthermore, it is also possible for other aromatic units to be incorporated by condensation, such as gallic acid, aminobenzenesulfonic acid, sulfanilic acid, phenolsulfonic acid, aniline, ammoniobenzoic acid, dialkoxybenzenesulfonic acid, dialkoxybenzoic acid, pyridine, pyridinemonosulfonic acid, pyridinedisulfonic acid, pyridinecarboxylic acid and pyridinedicarboxylic acid. Examples of suitable sulfonated  $\beta$ -naphthalene-formaldehyde condensates are the Melcret 500 L products distributed by Master Builders Solutions Deutschland GmbH.

(139) Generally, phosphonate containing dispersants incorporate phosphonate groups and polyether side groups.

(140) Suitable phosphonate containing dispersants are those according to the following formula  $R-(\text{OA}.\text{sup.2}).\text{sub.n6}-N-[CH.\text{sub.2}-PO(\text{OM}.\text{sup.3}.\text{sub.2}).\text{sub.2}].\text{sub.2}$  wherein R is H or a hydrocarbon residue, preferably a C.<sub>sub.1</sub>-C.<sub>sub.15</sub> alkyl radical, A.<sup>sup.2</sup> is independently C.<sub>sub.2</sub>-C.<sub>sub.18</sub> alkylene, preferably ethylene and/or propylene, most preferably ethylene, n<sub>6</sub> is an integer from 5 to 500, preferably 10 to 200, most preferably 10 to 100, and M.<sup>sup.3</sup> is H, an alkali metal, ½ alkaline earth metal and/or an amine.

(141) Preferably, the construction composition comprises less than 5 wt.-%, more preferably less than 3.5 wt.-%, most preferably less than 2 wt.-% of cementitious hydration products, relative to the total weight of the construction composition. It generally suffices to assess the following cementitious hydration products: ettringite, portlandite, syngenite. The presence and concentrations of these cementitious hydration products can be determined by Rietveld refinement of an X-ray diffraction (XRD) powder pattern. This means that the construction composition has no history of storage in high humidity environments. We believe that otherwise, ettringite among other cementitious hydration products is formed already in the powdery composition. Although these ettringite crystals are broken up at the time of mixing the construction composition with water at the time of use, the ettringite formation control provided by the invention is less prominent. Thus, storage of the construction composition in high humidity environments should be avoided.

(142) The invention also relates to the construction composition according to the invention in freshly mixed form, i.e. comprising water. Preferably, the ratio of water to cementitious binder a) is in the range of 0.2 to 0.7, preferably in the range of 0.25 to 0.5.

(143) The freshly mixed construction composition can be for example concrete, mortar or grout. The term “mortar” or “grout” denotes a cement paste to which are added fine aggregates, i.e. aggregates whose diameter is between 150 µm and 5 mm (for example sand), and optionally very fine aggregates. A grout is a mixture of sufficiently low viscosity for filling in voids or gaps. Mortar viscosity is high enough to support not only the mortars own weight but also that of masonry placed above it. The term “concrete” denotes a mortar to which are added coarse aggregates, i.e. aggregates with a diameter of greater than 5 mm.

(144) The construction composition may be provided as a dry mix to which water is added on-site to obtain the freshly mixed construction composition. Alternatively, the construction composition may be provided as a ready-mixed or freshly mixed composition.

(145) The aqueous construction composition is obtainable by mixing a powdery component C, containing the cementitious binder a) and the sulfate source c), and a liquid aqueous component W, wherein the ettringite formation controller d) is contained in one or both of components C and W. The polyol e) is preferably comprised in component W. The optional extraneous aluminate source b) is preferably comprised in component C.

(146) The sequence of addition of the optional ingredient f), i.e. at least one of a latent hydraulic binder, a pozzolanic binder and a filler material, depends primarily on the water content of ingredient f). When ingredient f) is provided in an essentially anhydrous form, it can conveniently be included in component C. Otherwise, and more commonly, ingredient f) is pre-mixed with component W, and component C is blended in subsequently.

(147) This mixing regimen prevents the immediate formation of ettringite, which would occur if the cementitious binder a) is exposed to water without the simultaneous presence of ettringite formation controller d).

(148) In a practical embodiment, the ettringite formation controller d), the polyol e) are dissolved in a part of the mixing water, and moist ingredients f), such as sand, are admixed. Subsequently, a pre-blended mix of the cementitious binder a), the sulfate source c), optionally the extraneous aluminate source b) and optionally anhydrous ingredients f), such as limestone, is added to the mixture. The remainder of the water is then added to adjust consistency.

(149) Preferably, the at least one of a latent hydraulic binder, a pozzolanic binder and a filler material f) is present in an amount of 500 to 1900 kg per m.<sup>sup.3</sup>, preferably 700 to 1700 kg per



m.sup.3, of the freshly mixed construction composition.

(150) The construction composition according to the invention is useful in applications such as producing building products, in particular for concretes such as on-site concrete, finished concrete parts, manufactured concrete parts (MCP's), pre-cast concrete parts, concrete goods, cast concrete stones, concrete bricks, in-situ concrete, ready-mix concrete, air-placed concrete, sprayed concrete/mortar, concrete repair systems, 3D printed concrete/mortar, industrial cement flooring, one-component and two-component sealing slurries, slurries for ground or rock improvement and soil conditioning, screeds, filling and self-levelling compositions, such as joint fillers or self-levelling underlayments, high performance concrete (HPC) and ultra high performance concrete (UHPC), hermetic fabricated concrete slabs, architectural concrete, tile adhesives, renders, cementitious plasters, adhesives, sealants, cementitious coating and paint systems, in particular for tunnels, waste water drains, screeds, mortars, such as dry mortars, sag resistant, flowable or self-levelling mortars, drainage mortars and concrete, or repair mortars, grouts, such as joint grouts, non-shrink grouts, tile grouts, injection grouts, wind-mill grouts (wind turbine grouts), anchor grouts, flowable or self-levelling grouts, ETICS (external thermal insulation composite systems), EIFS grouts (Exterior Insulation Finishing Systems, swelling explosives, waterproofing membranes or cementitious foams.

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## Description

### EXAMPLES

- (1) The invention is further illustrated by the appended drawing and the examples that follow.
- (2) FIG. 1 shows a plot of the photo current signal in mV against the time of dosage of CaCl.sub.2 in the calcium aluminate precipitation test according to one embodiment of the invention.

### METHODS

- (3) Testing Procedure—Open Time
- (4) Open time was determined with a Vicat needle according to DIN EN 480.
- (5) Calcium Aluminate Precipitation Test
- (6) For the calcium aluminate precipitation test, an automated titration module (Titrand 905, available from Metrohm) equipped with a high performance pH-electrode (iUnitrode with Pt 1000, available from Metrohm) and a photosensor (Spectrosense 610 nm, available from Metrohm) was used. First, a solution of 400 mL of a 1 wt.-% aqueous solution of a polyol to be investigated and 20 mL of a 1 mol/L NaOH aqueous solution was equilibrated for 2 min under stirring in the automated titration module. Then, 50 mL of a 25 mmol/L NaAlO.sub.2 aqueous solution was added thereto, followed by equilibration for another 2 min, obtaining an essentially clear test solution. In a next step, the test solution is titrated with a 0.5 mol/L CaCl.sub.2 aqueous solution which is dosed with a constant rate of 2 mL/min. During the whole experiment, the temperature is held constant at 20° C. The elapsed time to a turbidity inflection is recorded. To this end, the photo current signal in mV is plotted against the time of dosage of the CaCl.sub.2 aqueous solution. From the diagram, the onset point is determined as the intersection of the baseline tangent with a tangent to the curve after the inflection of the curve.
- (7) The invention is illustrated by the following examples. All wt.-% are understood as % bwoc, i.e., as relative to the mass of cementitious binder a). Various mortar mixes were prepared, adjusted to the same slump and their early strength development was measured. The basic recipe is as follows, to which further ingredients were added as described in detail below.
- (8) TABLE-US-00001 Material Amount [kg/m.sup.3] Cementitious binder 542 Limestone powder 68 Anhydrite (CAB 30) 54 Water 209 Quartz sand (0.1-0.3 mm) 155 Quartz sand (0.3-1 mm) 118 Natural sand (0-4 mm) 977 Crushed stones (2-5 mm) 279
- (9) Throughout examples 1 to 39, retarder 7 of WO 2019/077050 was used as glyoxylic acid urea

polycondensate (GA UC).

(10) In examples 40 to 45, the glyoxylic acid bisulfite adduct (GA BA) was used, the production of which is described in WO 2017/212045, additive 1.

(11) Mixing Procedure

(12) The crushed stones were dried in an oven at 70° C. for 50 h. Sands were dried in an oven at 140° C. for 68 h. Afterwards, the crushed stones and sands were stored at 20° C. for at least 2 d at 65% relative humidity. A glyoxylic acid urea polycondensate, sodium gluconate, NaHCO.sub.3 and a polycarboxylate based superplasticizer (Master Suna SBS 8000, available from Master Builders Solutions Deutschland GmbH) were added to the total amount of mixing water, to obtain a liquid aqueous component. Subsequently, crushed stones, sands, cementitious binder, anhydrite (CAB 30, available from Lanxess) and limestone were added to a 5 L Hobbart mixer. The liquid aqueous component was added thereto and the mixture was stirred for 2 min at level 1 (107 rpm) and for further 2 min at level 2 (198 rpm) to obtain an aqueous construction composition.

(13) Testing Procedure—Mini-Slump

(14) The used procedure is analogous to DIN EN 12350-2, with the modification that a mini-slump cone (height: 15 cm, bottom width: 10 cm, top width: 5 cm) was used instead of a conventional Abrams cone. 2 L of the aqueous construction composition were filled into the mini-slump cone. The cone was filled completely immediately after mixing. Afterwards, the cone was placed on a flat surface, and lifted, and the slump of the mortar mix was measured. The slump of all mixes was adjusted to 11 cm by adjusting the dosage of the superplasticizer to allow for comparability. The dosage of the superplasticizer was in the range of 0.11 to 0.22 wt.-% (dosage calculated as active substance).

(15) Testing Procedure—Early Strength Development

(16) The adjusted mortar mixes were each filled into mortar steel prisms (16/4/4 cm), and after 3 h at a temperature of 20° C. and relative humidity of 65%, a hardened mortar prism was obtained.

(17) The hardened mortar prism was demolded and compressive strength was measured according to DIN EN 196-1.

Reference Example: Calcium Aluminate Precipitation-Inhibiting Properties of Polyols

(18) Various polyols were assed for their precipitation-properties in the calcium aluminate precipitation test. The results are shown in the table that follows. For the control, 400 mL of bidestilled water was used instead of 400 mL of a 1 wt.-% aqueous solution of a polyol. The titration endpoint, expressed as the maximum calcium concentration (as Ca.sup.2+) before the onset of turbidity, is calculated from the elapsed time to the onset point. FIG. 1 shows a plot of the photo current signal in mV against the time of dosage of CaCl2. Curve a) of FIG. 1 shows the results in the absence of a polyol (“blank”). Curve b) of FIG. 1 shows the results for addition of 1% of triethanolamine. For curve b), a first tangent 1, referred to as “baseline tangent”, and a second tangent 2 are shown. From the baseline tangent 1 and the second tangent 2, the onset point in s may be determined as the intersection of the baseline tangent 1 with the second tangent 2.

(19) TABLE-US-00002 control (without ethylene triethanol- Polyol polyol) glycol glycerol amine erythrit Onset point [s] 42 42 64 500 686 Ca endpoint 59 59 93 682 924 [ppm]

Preparation and Evaluation of Construction Compositions

(20) Examples 1 to 15 (table 1) illustrate the impact of the total amount of available aluminate, i.e. available aluminate from the cementitious binder and the optionally present extraneous aluminate source, as well as the impact of glycerol, on early strength development. For examples 1 to 15, the following specifications apply: ratio of water to cementitious binder=0.37 NaHCO.sub.3=0.37 wt.-% sodium gluconate=0.077 wt.-% glyoxylic acid urea polycondensate=0.4 wt.-% (calculated as active substance) anhydrite=10 wt.-%

(21) The amount of available aluminate in the cementitious binder was determined by Rietveld refinement of an X-ray diffraction (XRD) powder pattern. Only the mineral phases C3A and C4AF were assessed.

(22) TABLE-US-00003 TABLE 1 total mol Al(OH)<sub>3</sub>— per ratio of compressive cementitious 100 g of cementitious Al(OH)<sub>3</sub>— glycerol strength after # binder binder to (SO<sub>2</sub>)<sub>2</sub>— [wt.-%] 3 h [MPa] 1 Colacem Gubbio 0.076 0.77 0.5 4 CEM | 2.5 R 2 Colacem Gubbio 0.076 0.77 1.5 16 CEM | 52.5 R 3 Couvrot 0.084 0.84 0.5 5 CEM | 52.5 R 4 Couvrot 0.084 0.84 1.0 0 20 CEM | 52.5 R 5 Mergelstetten 0.088 0.87 0.5 10 CEM | 42.5 N 6 .sup.[1] Mergelstetten 0.088 0.87 0.5 11 CEM | 42.5 N 7 Karlstadt 0.092 0.80 0.5 19 CEM | 42.5 R 8\* Gaurain 0.064 0.70 0 0 CEM | 52.5 R 9 Gaurain 0.064 0.70 0.5 2 CEM | 52.5 R 10 Gaurain 0.064 0.70 1.0 5 CEM | 52.5 R 11 Gaurain 0.083 .sup.[2] 0.91 1.0 7 CEM | 52.5 R 12 Gaurain 0.102 .sup.[3] 1.12 1.0 15 CEM | 52.5 R 13 Gaurain 0.102 .sup.[3] 1.12 1.5 21 CEM | 52.5 R 14 Aalborg White 0.060 .sup.[4] 0.62 1.5 5 CEM | 52.5 R 15\* Aalborg White 0.022 0.23 1.5 0 CEM | 52.5 R \*comparative example .sup.[1] The composition of example 6 did not comprise sodium gluconate. .sup.[2] = 0.064 mol aluminate/100 g cementitious binder + 0.019 mol additional aluminate/100 g cementitious binder from extraneous 1.5 wt.-% Al(OH)<sub>3</sub> .sup.[3] = 0.064 mol aluminate/100 g cementitious binder + 0.038 mol additional aluminate/100 g cementitious binder from extraneous 3.0 wt.-% Al(OH)<sub>3</sub> .sup.[4] = 0.022 mol aluminate/100 g cementitious binder + 0.038 mol additional aluminate/100 g cementitious binder from extraneous 3.0 wt.-% Al(OH)<sub>3</sub>

(23) Examples 16 to 26 (table 2) illustrate the impact of the molar ratio of total available aluminate to sulfate on early strength development. For examples 16 to 26, the following specifications apply: ratio of water to cementitious binder=0.37 NaHCO<sub>3</sub>=0.37 wt.-% sodium gluconate=0.077 wt.-% glyoxylic acid urea polycondensate=0.4 wt.-% (calculated as active substance) glycerol=0.5 wt.-%

(24) TABLE-US-00004 TABLE 2 total mol Al(OH)<sub>3</sub>— per ratio of compressive compressive 100 g of Al(OH)<sub>3</sub>— open strength strength cementitious cementitious anhydrite to time after 3 h after 24 h # binder binder [wt.-%] (SO<sub>2</sub>)<sub>2</sub>— [min] [Mpa] [Mpa] 16 Mergelstetten 0.088 5.0 1.11 n.d. 6 n.d. CEM | 42.5 N 17 Mergelstetten 0.088 10.0 0.76 n.d. 9 n.d. CEM | 42.5 N 18 Mergelstetten 0.088 12.5 0.66 n.d. 12 n.d. CEM | 42.5 N 19 Mergelstetten 0.088 15.0 0.58 n.d. 12 n.d. CEM | 42.5 N 20 Mergelstetten 0.088 20.0 0.46 n.d. 10 n.d. CEM | 42.5 N 21 Karlstadt 0.092 5.0 1.16 55 12 24 CEM | 42.5 R 22 Karlstadt 0.092 10.0 0.80 50 19 29 CEM | 42.5 R 23 Karlstadt 0.092 12.5 0.69 50 20 36 CEM | 42.5 R 24 Karlstadt 0.092 15.0 0.61 45 17 36 CEM | 42.5 R 25 Karlstadt 0.092 20.0 0.50 40 13 28 CEM | 42.5 R 26 Karlstadt 0.130 .sup.[1] 20.0 0.69 n.d. 21 (after 1 n.d. CEM | 42.5 R h) 26 (after 3 h) \* n.d. = not determined .sup.[1] = 0.092 mol aluminate/100 g cementitious binder + 0.038 mol additional aluminate/100 g cementitious binder from extraneous 3.0 wt.-% Al(OH)<sub>3</sub>

(25) Examples 27 to 30 (table 3) illustrate the impact of the ratio of water to cementitious binder on early strength development. For examples 27 to 30, the following specifications apply: NaHCO<sub>3</sub>=0.37 wt.-% sodium gluconate=0.077 wt.-% glyoxylic acid urea polycondensate=0.4 wt.-% (calculated as active substance) glycerol=2.0 wt.-% anhydrite=10 wt.-%

(26) TABLE-US-00005 TABLE 3 total mol Al(OH)<sub>3</sub>— per 100 g of ratio of water compressive cementitious cementitious to cementitious strength after # binder binder binder 3 h [Mpa] 27 Couvrot CEM | 52.5 R 0.085 0.37 20 28 Couvrot CEM | 52.5 R 0.085 0.30 33 29 Gaurain CEM | 52.5 R 0.064 0.37 9 30 Gaurain CEM | 52.5 R 0.064 0.30 22

(27) The open time for mortar mixes 1 to 30 was in each case at least 20 min. The open time may be further adjusted by incorporation of an α-hydroxy monocarboxylic acid salt, as is evident from the comparison of example 5 (open time 60 min) and example 6 (open time 30 min).

(28) For examples 31 to 37 (table 4), the following specifications apply: anhydrite (CAB 30)=15 wt.-% Al(OH)<sub>3</sub>—/(SO<sub>2</sub>)<sub>2</sub>—ratio=0.61 ratio of water to cementitious binder=0.37 glyoxylic acid urea polycondensate=0.5 wt.-% (calculated as active substance) Na<sub>2</sub>CO<sub>3</sub>=0.90 wt.-%

(29) TABLE-US-00006 TABLE 4 open compressive compressive cementitious dosage time strength after strength after # binder polyol [wt.-%] [min] 3 h [MPa] 24 h [MPa] 31 Karlstadt — —

60 3 n.d.\* CEM | 42.5 R 32 Karlstadt sucrose 0.3 60 8 10 CEM | 42.5 R 33 Karlstadt sucrose 0.9 30  
 12 n.d. CEM | 42.5 R 34\* Karlstadt ethylene 0.5 100 4 7 CEM | 42.5 R glycol 35 Karlstadt  
 triethanol- 0.5 65 16 20 CEM | 42.5 R amine 36 Karlstadt glycerol 0.5 70 15 22 CEM | 42.5 R 37  
 Karlstadt erythrit 0.5 65 20 26 CEM | 42.5 R n.d. = not determined \*comparative example  
 (30) Examples 38 to 45 (table 5) illustrate the impact of the carbonate source and the glyoxylic acid  
 bisulfite adduct on early strength development. For examples 38 to 45, the following specifications  
 apply: ratio of water to cementitious binder=0.37 ettringite formation controller=0.4 wt.-%,  
 calculated as active substance sodium gluconate=0.077 wt.-% glycerol=0.5 wt.-%  
 (31) TABLE-US-00007 TABLE 5 total mol Al(OH)<sub>3</sub>— per ratio of ettringite dosage  
 compressive compressive 100 g of Al(OH)<sub>3</sub>— formation dosage carbonate of d) open strength  
 strength cementitious cementitious anhydrite to controller of d) (i) source d) (ii) time after 3 h after  
 24 h # binder binder [wt.-%] (SO<sub>3</sub>)<sub>2</sub>— d) (i) [wt.-%] (ii) [wt.-%] [min] [MPa] [MPa] 38\*  
 Karlstadt 0.092 15.0 0.61 GA UC 0.4 — — 10 10 22 CEM | 42.5 R 39 Karlstadt 0.092 15.0 0.61  
 GA UC 0.4 propylene 0.37 95 15 23 CEM | 42.5 R carbonate 40 Karlstadt 0.092 5.0 1.16 GA BA  
 0.4 NaHCO<sub>3</sub> 0.37 25 11 19 CEM | 42.5 R 41 Karlstadt 0.092 10.0 0.80 GA BA 0.4  
 NaHCO<sub>3</sub> 0.37 20 16 28 CEM | 42.5 R 42 Karlstadt 0.092 15.0 0.61 GA BA 0.4  
 NaHCO<sub>3</sub> 0.37 20 13 22 CEM | 42.5 R 43 Karlstadt 0.092 5.0 1.16 GA BA + 0.36 +  
 NaHCO<sub>3</sub> 0.37 25 11 18 CEM | 42.5 R CA .sup.[1] 0.04 44 Karlstadt 0.092 10.0 0.80 GA BA  
 + 0.36 + NaHCO<sub>3</sub> 0.37 20 15 30 CEM | 42.5 R CA .sup.[1] 0.04 45 Karlstadt 0.092 15.0 0.61  
 GA BA + 0.36 + NaHCO<sub>3</sub> 0.37 15 17 24 CEM | 42.5 R CA .sup.[1] 0.04 .sup.[1] CA = citric  
 acid

## Claims

1. A construction composition comprising a) a cementitious binder comprising one or more calcium silicate mineral phases and one or more calcium aluminate mineral phases; b) optionally, an extraneous aluminate source; c) a sulfate source; d) an ettringite formation controller comprising (i) glyoxylic acid, a glyoxylic acid salt and/or a glyoxylic acid derivative; and (ii) at least one of a borate source and a carbonate source, wherein the carbonate source is selected from inorganic carbonates having an aqueous solubility of 0.1 g.Math.L.sup.-1 or more at 25° C., organic carbonates, and mixtures thereof; and e) a polyol in an amount of 0.2 to 2.5 wt.-%, relative to the amount of cementitious binder a); wherein the composition contains 0.05 to 0.2 mol of total available aluminate, calculated as Al(OH)<sub>3</sub>.sup.—, from the calcium aluminate mineral phases plus the optional extraneous aluminate source, per 100 g of cementitious binder a); and the molar ratio of total available aluminate to sulfate is 0.4 to 2.0.
2. The composition according to claim 1, wherein the calcium silicate mineral phases and calcium aluminate mineral phases constitute at least 90 wt.-% of the cementitious binder a), and the calcium silicate mineral phases constitute at least 60 wt.-% of the cementitious binder a).
3. The composition according to claim 1, wherein the calcium aluminate mineral phases are selected from C3A, C4AF and C12A7.
4. The composition according to claim 1, wherein the cementitious binder a) is Portland cement.
5. The composition according to claim 1, wherein the composition additionally comprises f) at least one of a latent hydraulic binder, a pozzolanic binder and a filler material.
6. The composition according to claim 1, wherein the optional extraneous aluminate source b) is selected from non-calciferous aluminate sources and calciferous aluminate sources.
7. The composition according to claim 1, wherein the sulfate source c) is a calcium sulfate source.
8. The composition according to claim 1, wherein the amount of polyol e) is 0.2 to 1 wt.-%, relative to the amount of cementitious binder a), if the Blaine surface area of the cementitious binder a) is 1500 to 4000 cm.sup.2/g.
9. The composition according to claim 1, wherein the polyol, in a calcium aluminate precipitation

test in which a test solution, obtained by supplementing 400 mL of a 1 wt.-% aqueous solution of the polyol with 20 mL of a 1 mol/L NaOH aqueous solution and 50 mL of a 25 mmol/L NaAlO<sub>2</sub> aqueous solution, is titrated with a 0.5 mol/L CaCl<sub>2</sub> aqueous solution at 20° C., inhibits precipitation of calcium aluminate up to a calcium concentration of 75 ppm.

10. The composition according to claim 9, wherein the polyol is selected from monosaccharides, oligosaccharides, water-soluble polysaccharides, compounds of general formula (P-I) or dimers or trimers of compounds of general formula (P-I):  $\text{---R---(CH}_2\text{---O---CH}_2\text{---O---CH}_2\text{---O---)}_n\text{---R---}$  wherein X is  $\text{---CH}_2\text{---}$  wherein R is  $\text{---CH}_2\text{OH}$ ,  $\text{---NH}_2$ , n is an integer from 1 to 4, m is an integer from 1 to 8.

11. The composition according to claim 1, wherein the glyoxylic acid derivative is a glyoxylic acid polymer.

12. The composition according to claim 1, wherein the inorganic carbonate is selected from sodium carbonate, sodium bicarbonate, potassium carbonate, lithium carbonate and magnesium carbonate; and the organic carbonate is selected from ethylene carbonate, propylene carbonate and glycerol carbonate.

13. The composition according to claim 1, wherein the ettringite formation controller additionally comprises (iii) a polycarboxylic acid or a salt thereof whose milliequivalent number of carboxyl groups is 3.0 meq/g or higher, if all the carboxyl groups are in unneutralized form.

14. The composition according to claim 13, wherein the polycarboxylic acid is selected from phosphonoalkyl carboxylic acids, amino carboxylic acids, and polymeric carboxylic acids.

15. The composition according to claim 1, wherein the ettringite formation controller additionally comprises (iv) a  $\alpha$ -hydroxy monocarboxylic acid or a salt thereof.

16. The composition according to claim 1 additionally comprising a dispersant selected from comb polymers having a carbon-containing backbone to which are attached pendant cement-anchoring groups and polyether side chains, non-ionic comb polymers having a carbon-containing backbone to which are attached pendant hydrolysable groups and polyether side chains, the hydrolysable groups upon hydrolysis releasing cement-anchoring groups, colloidally disperse preparations of polyvalent metal cations, such as Al<sup>3+</sup>, Fe<sup>3+</sup> or Fe<sup>2+</sup>, and a polymeric dispersant which comprises anionic and/or anionogenic groups and polyether side chains, and the polyvalent metal cation is present in a superstoichiometric quantity, calculated as cation equivalents, based on the sum of the anionic and anionogenic groups of the polymeric dispersant, sulfonated melamine-formaldehyde condensates, lignosulfonates, sulfonated ketone-formaldehyde condensates, sulfonated naphthalene-formaldehyde condensates, phosphonate containing dispersants, phosphate containing dispersants, and mixtures thereof.

17. The composition according to claim 1, wherein the construction composition comprises less than 5 wt.-% of cementitious hydration products, relative to the total weight of the construction composition.

18. The construction composition according to claim 1, in freshly mixed form, wherein the ratio of water to cementitious binder a) is in the range of 0.2 to 0.7.

19. The composition according to claim 1, wherein the amount of polyol e) is 1 to 2.5 wt.-%, relative to the amount of cementitious binder a), if the Blaine surface area is more than 4000 cm<sup>2</sup>/g.

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